



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME
DSA0115 Object Oriented Programming with C++
for System Level Programming



LEVEL 1 QUESTIONS

1. Write a program to check whether a number is a **Harshad number** (divisible by the sum of its digits).

QUESTIONS 33 / 48 completed

Harshad

HARSHAD

Write a program to read in an integer and determine if it is a Harshad number or not.

PROGRAM EDITOR

C++ Run Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int n, temp, rem, sum = 0;
7
8     cin >> n;
9     temp = n;
10
11     while(n > 0)
12     {
13         rem = n % 10;
14         sum = sum + rem;
15         n = n / 10;
16     }
17
18     if(temp % sum == 0)
19         cout << "Harshad Number";
20     else
21         cout << "Not Harshad Number";
```

OUTPUT

Harshad Number

INPUT

18

2. Swap two numbers **without** using a temporary variable.

QUESTIONS 33 / 48 completed

Swap

SWAP

Write a program to read n two integers and swap their values using the bitwise XOR operator.

PROGRAM EDITOR C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,b;
5     cin>>a>>b;
6     a=a^b;
7     b=a^b;
8     a=a^b;
9     cout<<a<<" "<<b;
10    return 0;
11 }
```

OUTPUT

```
10 5
```

INPUT

```
5
10
```

3. Hollow square pattern.

main.cpp

Run

Clear

```
1 #include <stdio.h>
2 int main() {
3     int n, i, j;
4     printf("Enter size: ");
5     scanf("%d", &n);
6     for(i = 1; i <= n; i++) {
7         for(j = 1; j <= n; j++) {
8             if(i == 1 || i == n || j == 1 || j == n)
9                 printf("* ");
10            else
11                printf(" ");
12        }
13        printf("\n");
14    }
```

Output

```
Enter size: 5
* * * * *
*   *
*   *
*   *
* * * * *
```

=== Code Execution Successful ===

4. Number pattern.

Programiz C++ Online Compiler

Programiz PRO

main.cpp

Run

Output

Clear

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n;
5     cout << "Enter number of rows: ";
6     cin >> n;
7     for (int i = 1; i <= n; i++) {
8         for (int j = 1; j <= i; j++) {
9             cout << j << " ";
10        }
11        cout << endl;
12    }
13    return 0;
14 }
```

Enter number of rows: 6

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

1 2 3 4 5 6

=== Code Execution Successful ===

5. Check whether a number is a **palindrome**.

QUESTIONS 33 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n, temp, rev = 0, rem;
6     cin >> n;
7     temp = n;
8     while(n > 0)
9     {
10         rem = n % 10;
11         rev = rev * 10 + rem;
12         n = n / 10;
13     }
14     if(temp == rev)
15         cout << "Palindrome";
16     else
17         cout << "Not Palindrome";
18     return 0;
19 }
```

OUTPUT

Palindrome

INPUT

101

6. Find the sum of the cubes of the first n natural numbers.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 int main() { 4 int n; 5 long long sum = 0; 6 cout << "Enter n: "; 7 cin >> n; 8 for (int i = 1; i <= n; i++) { 9 sum += i * i * i; 10 } 11 cout << "Sum of cubes = " << sum; 12 return 0; 13 }</pre>	<pre>Enter n: 10 Sum of cubes = 3025 === Code Execution Successful ===</pre>

7. Convert binary to decimal.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 int main() { 4 int binary, decimal = 0, base = 1, rem; 5 cout << "Enter a binary number: "; 6 cin >> binary; 7 while (binary > 0) { 8 rem = binary % 10; 9 decimal += rem * base; 10 binary /= 10; 11 base *= 2; } 12 cout << "Decimal = " << decimal; 13 return 0; 14 }</pre>	<pre>Enter a binary number: 1011 Decimal = 11 === Code Execution Successful ===</pre>

8. Palindrome pattern.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 int main() { 4 int n; 5 cout << "Enter number of rows: "; 6 cin >> n; 7 for (int i = 1; i <= n; i++) { 8 for (int j = i; j >= 1; j--) 9 cout << j; 10 for (int j = 2; j <= i; j++) 11 cout << j; 12 cout << endl; } 13 return 0; 14 }</pre>	<pre>Enter number of rows: 6 1 212 32123 4321234 543212345 65432123456 === Code Execution Successful ===</pre>

9. Hollow pyramid pattern.

main.cpp

Run

Output

Clear

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n;
5     cout << "Enter number of rows: ";
6     cin >> n;
7     for (int i = 1; i <= n; i++) {
8         for (int j = 1; j <= n - i; j++)
9             cout << " ";
10        for (int j = 1; j <= 2 * i - 1; j++) {
11            if (i == n || j == 1 || j == 2 * i - 1)
12                cout << "**";
13            else
14                cout << " ";
15        }
16        cout << "\n";
17    }
18 }
```

Enter number of rows: 6

```
      *
     **
    ***
   ****
  *****
 *****
```

=== Code Execution Successful ===

10. Find the sum of the squares of the first n natural numbers.

Programs

On-Going Compiler

main.cpp

Run

Output

Clear

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n;
5     long long sum = 0;
6     cout << "Enter n: ";
7     cin >> n;
8     for (int i = 1; i <= n; i++) {
9         sum += i * i;
10    }
11    cout << "Sum of squares = " << sum;
12    return 0;
13 }
```

Enter n: 8
Sum of squares = 204

=== Code Execution Successful ===